FLDA

Objective fct.:

maximizes this quantity

$$J(w) = \frac{w^T S_B \, w}{w^T S_W \, w}$$

where $S_B \equiv$ between-class scatter matrix

$S_W \equiv$ within-class scatter matrix.

optimizing for $J(w)$:

$$\frac{\partial J}{\partial w} = 0$$

$\iff$ $S_B \cdot w = \lambda \cdot S_W \cdot w$

where $\lambda = \dfrac{w^T \cdot S_B \cdot w}{w^T \cdot S_W \cdot w} \in \mathbb{R}$

$\iff$ $(S_W^{-1} \cdot S_B) \, w = \lambda \cdot w$

if S_W^{-1}

$\therefore$ w is an eigenvector of $(S_W^{-1} \cdot S_B)$.

with Largest eigenvalue λ.

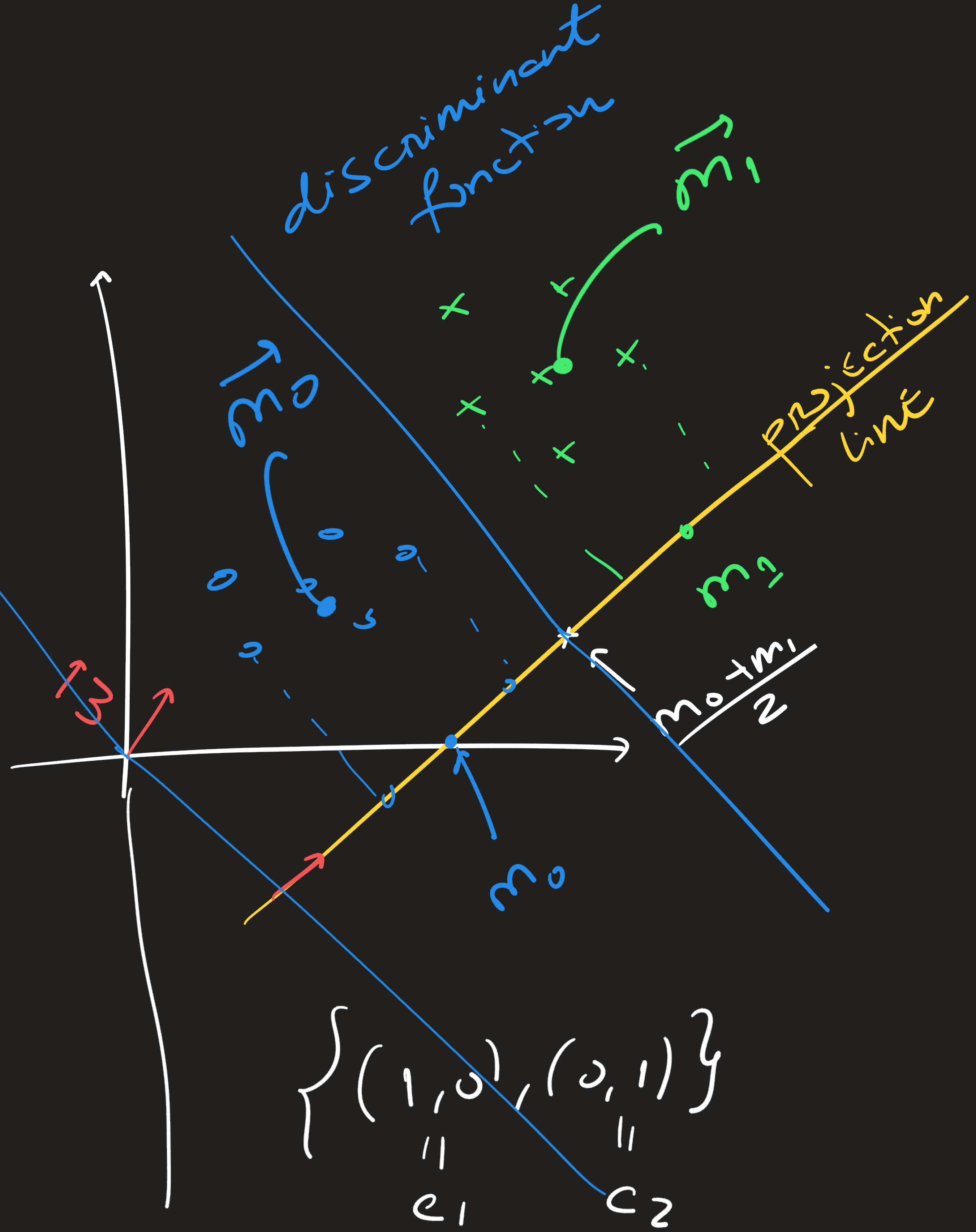
$$w \leftarrow \frac{w}{\|w\|}$$

this ensures it has length 1.

Mapper fct:

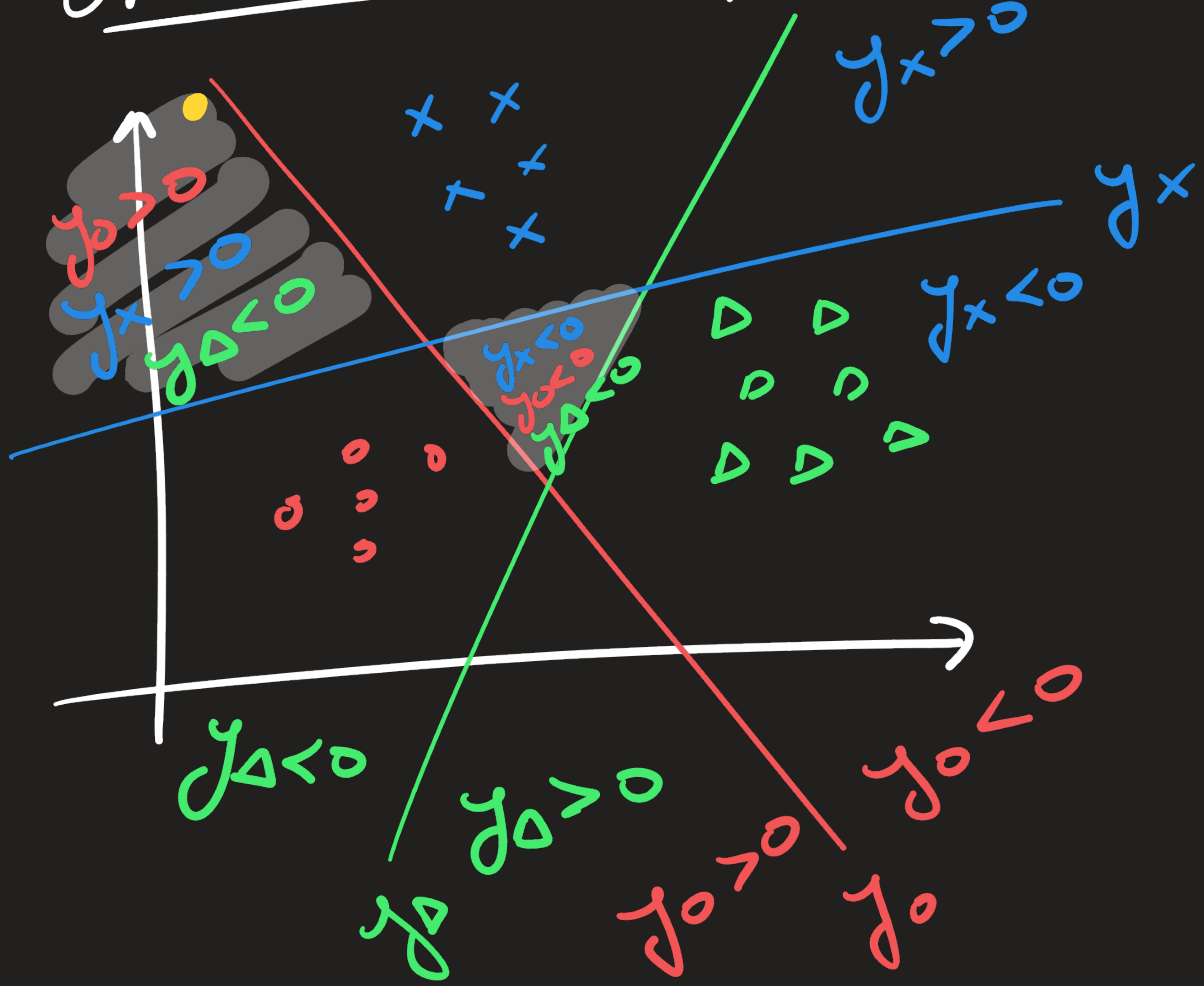
$$y(x) = w^T x + w_0$$

Eigenvector of $S_W^{-1} \cdot S_B$ w/ largest eigenvalue



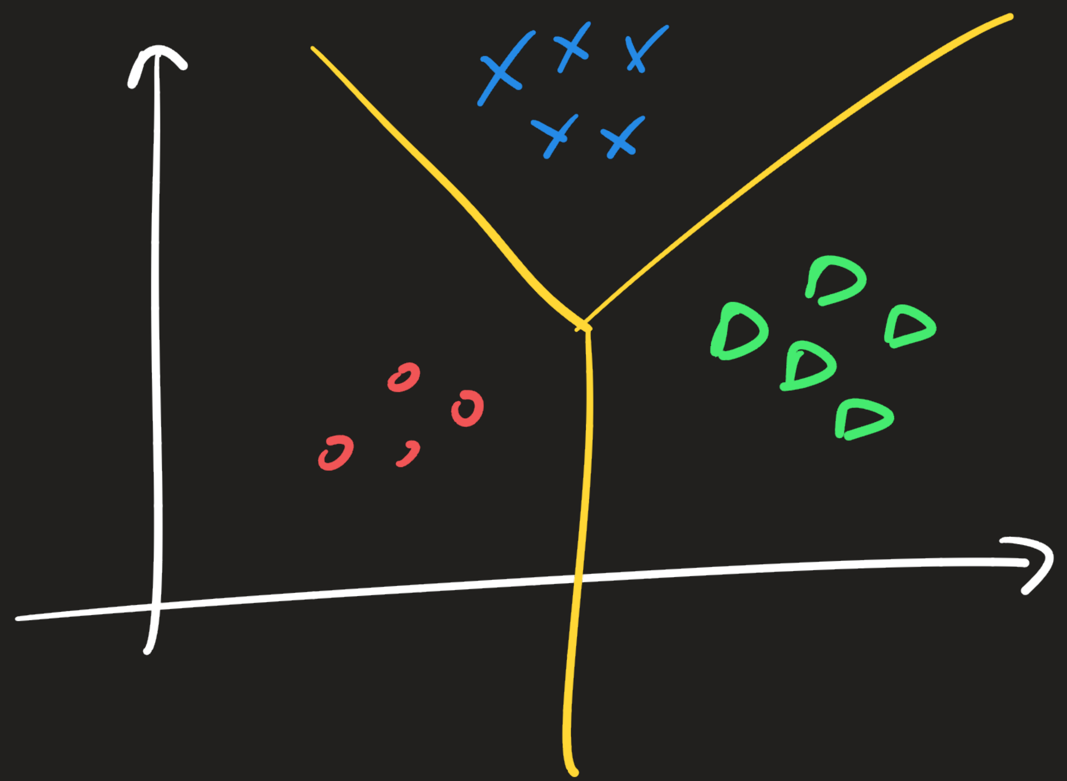
$$w_0 = -\left(\frac{\vec{m_0}+\vec{m_1}}{2}\right)^T \cdot \vec{w}$$

ONE-VS-All Approach

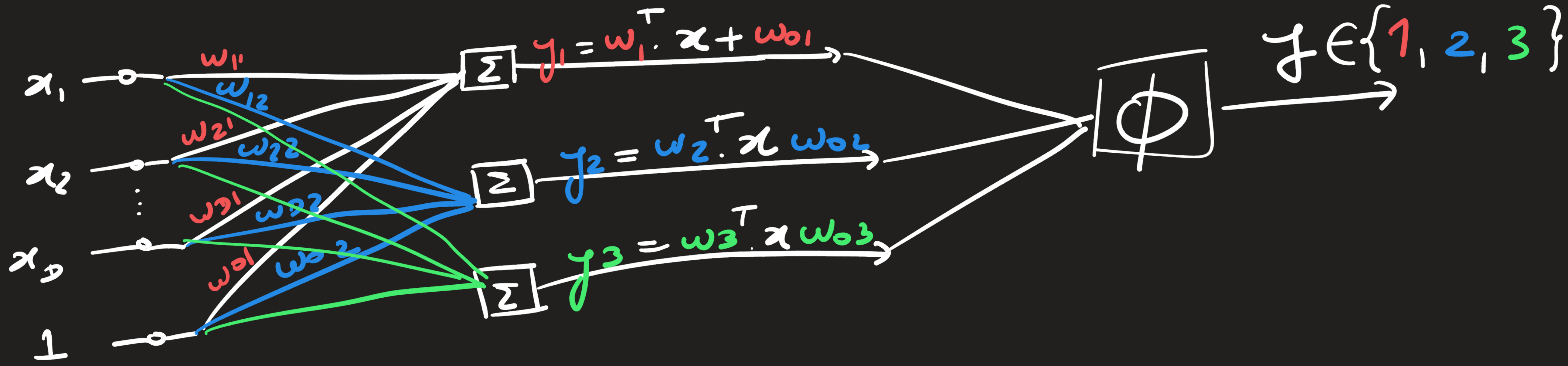


$$y_i = w_i^T x + w_{0i} \equiv \text{discriminant fct. for } C_i$$

$$\boxed{x \in \text{Class } j \text{ if } y_j(x) > y_k(x), \; \forall k \neq j}$$



<u>Singly convex</u>
<u>Regions.</u>



input sample of D-dimensions + bias input

$$\phi = \arg\max_j([y_1, y_2, y_3])$$